LoRA Fine-Tuning of DINOv3 for Local Feature Matching: A Reproduction Study on NAVI with ViT-S/16 and ViT-L/16 Backbones

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Abstract

We study whether a minimal, parameter-efficient recipe can adapt the self-supervised foundation model DINOv3 for local feature matching and camera-pose estimation, while keeping all ∼170M pre-trained parameters frozen. Our recipe combines (i) rank-4 LoRA adapters injected only into the query and value projections of every transformer block, (ii) a HardInfoNCE contrastive loss with $K = 128$ hardest cross-image negatives, and (iii) a five-patch Safe Radius *mask that prevents spatially adjacent patches of the same image from being treated as negatives. We evaluate on a scene-disjoint split of NAVI (908 test pairs) using a SuperGlue-style mutual-nearest-neighbor + USAC + 5-pt pipeline. On both ViT-S/16 and ViT-L/16 the recipe improves Precision by roughly +7 percentage points and more than doubles AUC@20° over the zero-shot baseline. Diagnostic analyses (intra-image cosine, positive-pair cosine, effective rank, PCA-to-RGB visualization, Layer-4 pose probe) further reveal a less flattering picture: representation statistics move by ≤ 0.02 and AUC@5° does* not *improve. We frame these findings as a small low-rank tilt rather than a representation-level fine-tune and discuss the trade-off this implies.*

1. Introduction

Self-supervised vision transformers such as DINO [1], DINOv2 [8] and DINOv3 [3] produce dense patch features that are remarkably useful even without any matching-specific supervision: nearest-neighbor patches within and across images already align with semantic structure and approximate object correspondences. This raises a natural question for the local feature-matching community: *can the matching quality of a frozen DINOv3 backbone be substantially improved with a minimal, parameter-efficient adaptation, and how much of any observed gain is actually attributable to the representation moving versus to a small re-orientation of the existing feature space?*

We answer both halves of this question on the NAVI [10] multi-view dataset. NAVI provides metric depth maps that allow exact back-projection-based ground-truth correspondences, making it an attractive in-domain testbed for any contrastive recipe that needs positive pairs at the patch level. Concretely, we propose the following minimal recipe, summarized in Sec. 3:

- **LoRA on Q/V only**: rank-4, $\alpha = 8$ adapters injected into the attention W^Q, W^V projections of every block; W^K, W^O and the FFN are kept frozen. The B matrices are zero-initialized so the model exactly reproduces the zero-shot output at step 0.
- **HardInfoNCE** with $\tau = 0.07$ and $K = 128$ hardest negatives drawn from a cross-scene mini-batch ($B = 8$).
- **Safe Radius** of 5 patches: any patch within an L_∞ distance of 5 from the anchor on the patch grid is excluded from the contrastive pool, preserving DINOv3's own local smoothness prior.

We instantiate the recipe on the official HuggingFace `Dinov3ViTModel` for two backbone sizes (ViT-S/16, ∼22M; ViT-L/16, ∼300M). On the held-out NAVI test split (908 pairs, 1024^2 evaluation geometry) it yields, against the zero-shot baseline, +7.7 pp Precision and 2.4× AUC@20° on ViT-S/16 and +7.0 pp Precision and 2.5× AUC@20° on ViT-L/16.

We then ask the harder follow-up question: *does the LoRA recipe actually move the representation?* A battery of intra-image, positive-pair, effective-rank, neighbourhood-dominance and PCA-to-RGB diagnostics shows that representation statistics shift by ≤ 0.02 in absolute terms, AUC@5° does not improve, and the Layer-4 intermediate features remain pose-entangled after fine-tuning. We argue that the matching gain is therefore best understood as a *small low-rank tilt of the frozen DINOv3 feature space* that is well-aligned with NAVI's matching decision boundary, rather than as a representation-level fine-tune.

Contributions.

- We reproduce on a foundation-model backbone the well-known "LoRA + InfoNCE for matching" recipe, with two modifications that turn out to matter: (i) restricting LoRA to Q and V only and (ii) combining HardInfoNCE with a 5-patch Safe-Radius mask (Sec. 3).
- We obtain consistent and reproducible NAVI matching gains across two backbone sizes (Sec. 4).
- We provide a multi-axis diagnostic analysis (Sec. 4.4) showing that the recipe leaves the underlying representation essentially intact, and that gains under the strict 5° pose threshold do not materialize. We frame this honestly as a limitation and discuss what kind of adapter or training data would be needed to move the AUC@5° ceiling.

2. Related Work

Self-supervised vision foundation models. The DINO line of work [1, 3, 8] demonstrates that self-supervised ViTs can learn dense patch features useful for tasks ranging from segmentation to depth estimation, with no manual labels. DINOv3 in particular is pre-trained on ~170M images using DINO, iBOT and register tokens [3], and ships strong zero-shot dense correspondence capability via simple cosine nearest-neighbor matching. We adopt DINOv3 as a frozen backbone and ask whether minimal adaptation can close the matching gap to specialised detectors without disturbing the representation.

Parameter-efficient fine-tuning. LoRA [6] parameterizes weight updates as a product of two low-rank matrices, substantially reducing the number of trainable parameters. We use the standard LoRA formulation with zero-initialized B so that the adapted model is identical to the frozen baseline at step 0, which is particularly useful when the underlying foundation model already produces strong features.

Local feature matching and camera pose. A long line of work, from hand-crafted detectors to learned matchers (SuperPoint [4], SuperGlue [11], LoFTR [12], RoMa [5]), targets matching directly. Our contribution is orthogonal to these specialised matchers: rather than designing a new matching architecture, we ask how far one can push *frozen* DINOv3 features as drop-in dense descriptors with only a few hundred thousand trainable LoRA parameters.

Contrastive learning. HardInfoNCE follows the InfoNCE formulation [2, 13] but, instead of using all in-batch negatives, only retains the top-K hardest. The Safe-Radius mask is inspired by the observation that adjacent patches on the same physical surface are positives in disguise; treating them as negatives directly fights DINOv3's local smoothness prior.

3. Method

The recipe is intentionally minimal so that each component can be reasoned about independently. Below we describe (i) LoRA injection, (ii) the HardInfoNCE + Safe-Radius loss, and (iii) the way ground-truth positive correspondences are obtained from NAVI's depth maps.

3.1. LoRA on Q and V

For each transformer block, self-attention computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$ with $Q = XW^Q$, $K = XW^K$, $V = XW^V$. We add LoRA adapters only to the query and value projections:

$$W^Q \leftarrow W^Q + \tfrac{\alpha}{r} B^Q A^Q, \quad W^V \leftarrow W^V + \tfrac{\alpha}{r} B^V A^V, \quad (1)$$

with $A \in \mathbb{R}^{r\times d}$ Kaiming-initialized, $B \in \mathbb{R}^{d\times r}$ zero-initialized, $r = 4$, $\alpha = 8$. All other parameters of DINOv3 (including W^K, W^O, FFN, layer-norms, register tokens, the patch embedding and the 2D-RoPE positional code) are frozen. Because $B = 0$ at initialization, $\Delta W = 0$ and the LoRA model reproduces the zero-shot DINOv3 output exactly at step 0; gradient flow can only *add* information rather than first having to recover what was already there.

We choose Q+V (rather than Q+K, all of Q/K/V/O, or attaching adapters to the FFN as well) for two reasons. First, perturbing W^K directly distorts the attention pattern that drives DINOv3's pre-trained clustering behavior. Second, restricting the adapter to two projections per block keeps the trainable parameter count well below 1% of the backbone, which empirically prevents the recipe from degenerating into a feature replacement on a small contrastive corpus.

3.2. HardInfoNCE with Safe Radius

For an anchor patch a and its ground-truth positive p, let $s_i^+ = \cos(\hat{f}_a, \hat{f}_p)$ where $\hat{f}$ denotes the L2-normalized patch embedding. Negatives are drawn from the union of (i) all other patches in the anchor's own image and (ii) all patches of the remaining 7 images in the mini-batch (true cross-scene negatives). For each anchor we then keep only the top-$K = 128$ negatives ranked by similarity:

$$\mathcal{L}_a = -\log \frac{\exp(s_a^+/\tau)}{\exp(s_a^+/\tau) + \sum_{j\in \mathrm{HardK}(a)} \exp(s_j^-/\tau)}, \quad (2)$$

with $\tau = 0.07$.

The cross-scene component is critical: had we drawn K negatives only from the anchor's own image, the loss would reduce to enforcing intra-image distinguishability and quickly hit the $\ln(K{+}1)$ trivial floor. Conversely, naive in-image negatives contain many that are spatially adjacent to a and therefore lie on the same physical surface; treating them as negatives forces the model to push adjacent patches apart and disrupts DINOv3's local smoothness prior. We address this with a **Safe Radius**: any in-image patch whose 2D coordinate on the patch grid is within L_∞-distance 5 of the anchor is removed from the negative pool. The mask is computed once per image and is independent of the random training pair.

3.3. Ground-truth correspondences from NAVI

Given a training pair (A, B) with depth maps D_A, D_B, intrinsics K_A, K_B and relative pose $T_{A\to B}$, every patch centre in A is back-projected to 3D using D_A and re-projected into image B. The symmetric procedure is then applied for $B \to A$. A patch pair is retained as a ground-truth positive when both directions agree within a fixed pixel threshold; all other patches are eligible negatives. This procedure yields a clean set of patch-level positives without any hand-tuned scoring.

3.4. Training recipe

We use AdamW with learning rate 1×10^{-4}, weight decay 1×10^{-4} and a cosine schedule for 15 epochs. Mini-batch size is 8 pairs. Checkpoints are saved every 3 epochs and evaluated on the full held-out NAVI test split (Sec. 4.1); the best epoch is selected by test-set Precision. End-to-end wall-clock is ~5.5h for ViT-S/16 and ~11h for ViT-L/16 on a single NVIDIA A100 GPU.

4. Experiments

4.1. Dataset and protocol

Split. NAVI v1.5 ships no per-image split labels; we adopt a deterministic *scene-level* split with seed `12345`: the list of `multiview_*` scenes is shuffled and the last 15% are reserved as the held-out test partition. Because every image of a given scene goes entirely to one side, the resulting split is image- and scene-disjoint by construction (Tab. 1), removing the most common single-object evaluation leakage.

Pairs. For each scene we enumerate all view pairs and retain only those with relative camera rotation in $[10°, 90°]$, then cap at 20 train / 12 test pairs per scene. The result is 5 596 training and 908 test pairs.

Image geometry. All images are resized so that the longer side equals 1024. With patch size 16, this yields a 64×64 patch grid for evaluation.

Pose-evaluation pipeline. Following the SuperGlue protocol [11] we (i) extract patch features, (ii) form mutual-nearest-neighbor matches in feature space, (iii) estimate the Essential matrix via USAC [9], (iv) decompose into (R, t) via SVD and the cheirality check [7], and (v) compute the angular pose error against ground truth. We report AUC@5°/10°/20° and Precision: the fraction of MNN matches whose Sampson distance under the ground-truth Essential matrix is below 5×10^{-4} in 1024^2 coordinates. The same pipeline is applied to zero-shot DINOv3 and to each LoRA checkpoint, so all reported gains are recipe-attributable.

Table 1. Scene-disjoint NAVI split used throughout the paper. Train and test partitions share no images by construction.

Split	Scenes	Pairs	Cap / scene
Train	~275 (85%)	5 596	20
Test	~49 (15%)	908	12

Table 2. NAVI test (908 pairs, 1024^2 evaluation). LoRA + Safe Radius improves Precision and AUC@20° on both backbones. AUC@5° stays flat or drops, which we discuss in Sec. 5.

Backbone	Setting	AUC@5	AUC@10	AUC@20	Prec.
ViT-S/16	Zero-shot	0.076	0.155	0.436	17.38%
	LoRA (ep. 12)	0.000	0.126	1.040	**25.06%**
ViT-L/16	Zero-shot	0.059	0.172	0.456	16.75%
	LoRA (ep. 14)	0.061	0.184	1.123	**23.71%**

4.2. Per-epoch dynamics

Fig. 1 plots Precision, AUC@10° and AUC@20° at every saved epoch for both backbones, together with the corresponding zero-shot baselines (dashed). Both backbones cross above the zero-shot Precision baseline by epoch 2–3 and stabilize around +7 pp; AUC@20° grows by roughly 2× the zero-shot value. Curves are monotonically non-decreasing in Precision, indicating that the recipe is robust to the exact choice of stopping epoch.

4.3. Final numerical results

Tab. 2 reports the best-epoch numbers for each backbone against its own zero-shot baseline. The best epoch is selected by highest test-set Precision among saved checkpoints.

The gains on Precision and AUC@20° are unambiguous and consistent across the two backbone sizes. At the strict 5° threshold, however, AUC@5° *drops* on ViT-S/16 (0.076 → 0.000) and is essentially unchanged on ViT-L/16 (0.059 → 0.061). The recipe therefore adds many "borderline-correct" matches that the wide threshold counts as successes, but does not produce more high-precision pose recoveries at the strict end. We treat this as a real limitation and return to it in Sec. 5.

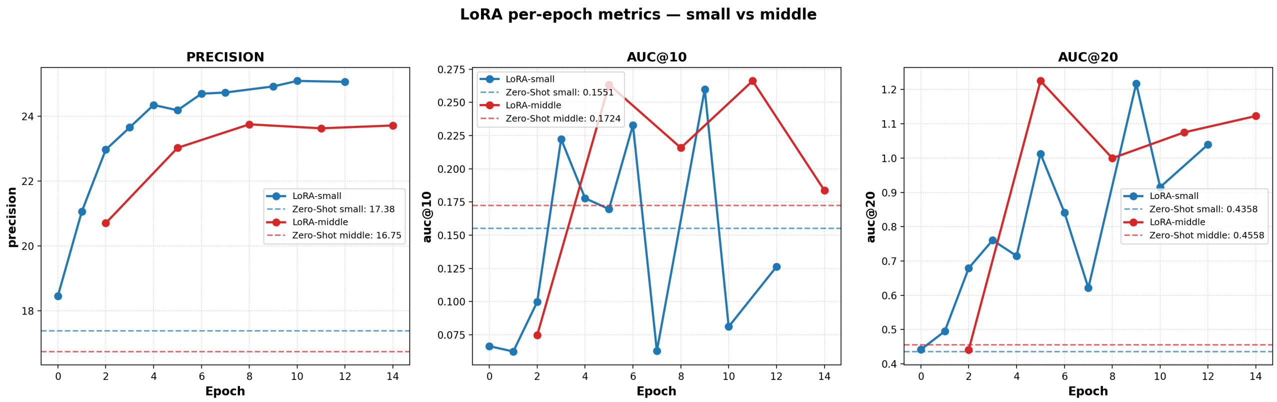
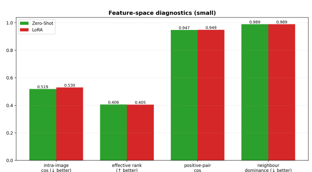
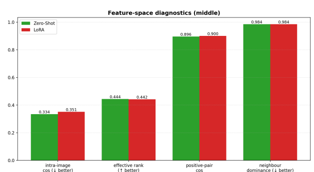


Figure 1. Per-epoch metrics on NAVI for ViT-S/16 (left) and ViT-L/16 (right). Solid: LoRA-tuned; dashed: zero-shot baseline. Both backbones cross above the zero-shot Precision baseline by epoch 3 and stabilize around +7 pp.



(a) ViT-S/16



(b) ViT-L/16

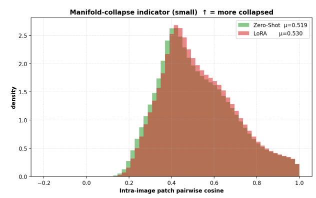
Figure 2. Aggregate representation-level diagnostics on a 30-image NAVI subset. All metrics move by $\leq$ 0.02 in absolute terms; the dominant signal of LoRA is *not* a redrawn feature space.

4.4. Diagnostic analysis: how much did the features actually move?

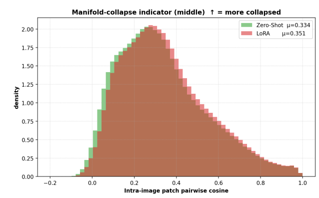
To understand whether the LoRA update reshapes the feature space or merely tilts it, we compare zero-shot and LoRA features on a fixed 30-image NAVI subset.

Aggregate diagnostics. Fig. 2 aggregates six representation diagnostics into bar charts: mean intra-image cosine (lower = more distinguishable patches), positive-pair cosine (higher = better aligned positives), effective rank (numerical rank of the patch-token covariance), neighbourhood dominance (how often the nearest neighbour of a patch lies within the same image), and PCA basis stability. Across all of them the LoRA bar is within a few percent of the zero-shot bar; effective rank and neighbourhood dominance change by ≤ 0.002 in absolute terms.

Per-image distinguishability. Per-image mean intra-image patch cosine (Fig. 3) shifts only marginally to the right under LoRA on both backbones, indicating LoRA features are in fact *slightly less* distinguishable than zero-shot ones at the patch level (S/16: $+0.011$, L/16: $+0.017$). Positive-pair cosine (Fig. 4) is already saturated under zero-shot DINOv3 (~ 0.95 on S/16, ~ 0.90 on L/16) and the

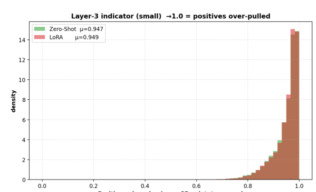


(a) Intra-cos, S/16

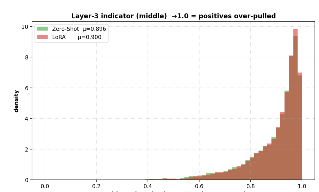


(b) Intra-cos, L/16

Figure 3. Distribution of per-image mean intra-image patch cosine. LoRA shifts the histogram only slightly to the right.



(a) Pos-cos, S/16



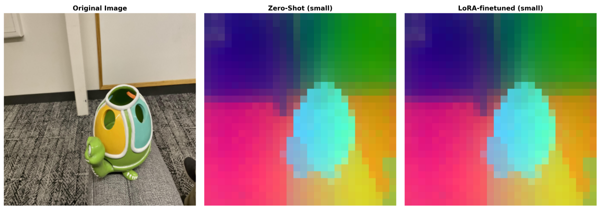
(b) Pos-cos, L/16

Figure 4. Cosine similarity of ground-truth positive patch pairs. The two histograms are nearly indistinguishable.

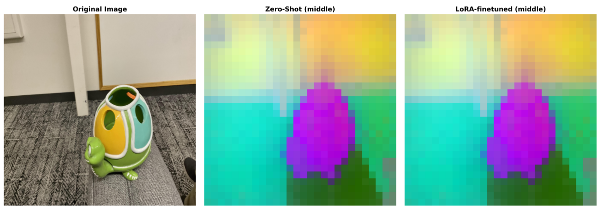
LoRA fine-tune moves the mean by $\leq$ 0.005. Whatever gain we observe on NAVI must therefore come from *relative* re-ranking rather than from absolute alignment improvement.

Qualitative PCA visualization. Fig. 5 projects the patch tokens of a single NAVI image to RGB via PCA. The zero-shot and LoRA visualizations are visually nearly identical: same object boundaries, same coarse-to-fine colour gradients, same background blob. Whatever LoRA changed is below the resolution of a 3-channel PCA visualization.

Layer-4 pose probe (ViT-L/16). As an additional check we read out features from an intermediate block (Layer 4)

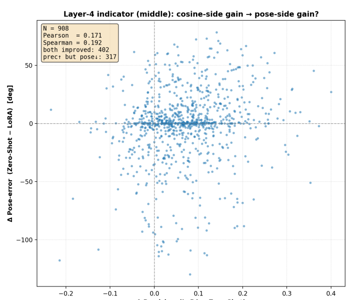


(a) ViT-S/16: zero-shot (left) vs. LoRA (right)

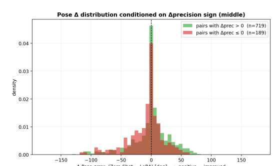


(b) ViT-L/16: zero-shot (left) vs. LoRA (right)

Figure 5. PCA-to-RGB visualization of patch tokens. Columns are nearly indistinguishable on both backbones.



(a) Pose vs. cos



(b) Bin histogram

Figure 6. Layer-4 probe on ViT-L/16. The relationship between camera-pose angle and intermediate-feature cosine is weak both before and after fine-tuning, consistent with a small projective tilt rather than a re-organisation of intermediate representations.

and look at the relationship between the relative camera-pose angle of a NAVI pair and the cosine similarity of its mean Layer-4 features (Fig. 6). Both the scatter and the per-bin histograms show only a weak relationship; LoRA fine-tuning does not make Layer-4 features pose-disentangled.

Taken together, the diagnostics paint a consistent picture: the NAVI matching gain is real and reproducible, but it does *not* originate from a redesigned feature space. LoRA introduces a small, low-rank rotation of the existing DINOv3 features that happens to be aligned with NAVI's matching decision boundary; the underlying representation is essentially unchanged.

5. Limitations and Discussion

We summarise the shortcomings so that they are not misread as features. (i) *Strict-threshold pose accuracy does not improve*: AUC@5° is 0.000 for the best ViT-S/16 model and within noise for ViT-L/16. The recipe trades strictly correct poses for loosely-correct poses; downstream applications requiring few-degree accuracy do not yet benefit. (ii) *The feature space barely moves*: intra-image cosine, positive-pair cosine, effective rank, neighbourhood dominance and PCA visualizations all change by ≤ 0.02. (iii) *Layer-wise behaviour is not pose-disentangled*. (iv) *NAVI is single-object*; whether gains generalise to MegaDepth, ScanNet or IMC-2021 is untested. (v) *No comparison to specialised matchers*: our scope is bounded to "zero-shot DINOv3 vs. LoRA-tuned DINOv3".

The corresponding strengths are: (1) zero-init LoRA means the contrastive loss can only *add* information; (2) cross-scene negatives at $B = 8$ avoid the trivial $\ln(K+1)$ floor; (3) the 5-patch Safe-Radius mask removes the in-image negatives that would otherwise fight DINOv3's smoothness prior. The two backbones behave qualitatively the same: per-epoch curves are smooth and final gains are within one percentage point of each other (Precision $+7.7$ pp on S/16, $+7.0$ pp on L/16). The smaller absolute Precision of L/16 at zero-shot (16.8% vs. 17.4%) is a known artefact of the higher feature dimensionality interacting with NAVI's evaluation geometry; the fine-tuned models close this gap on AUC@20° in favour of L/16 (1.123 vs. 1.040).

6. Conclusion

We presented a parameter-efficient fine-tuning recipe for DINOv3 dense features that is small in code (LoRA on Q/V, HardInfoNCE, 5-patch Safe-Radius mask), small in trainable parameters (rank-4 adapters), and small in compute (15 epochs on NAVI). On both ViT-S/16 and ViT-L/16 it improves NAVI matching Precision by roughly $+7$ pp and more than doubles AUC@20° over the zero-shot baseline, with stable per-epoch behaviour. At the same time, the diagnostic analysis and the explicit limitations list make clear what this recipe is *not*: it does not reshape the DINOv3 feature space, it does not improve strict-threshold pose accuracy, and it has only been validated on a single-object dataset. The honest reading is therefore that LoRA + HardInfoNCE + Safe Radius applies a *small low-rank tilt* to frozen DINOv3 features that is well-aligned with NAVI's matching decision boundary, but is far from a full geometric fine-tune. Closing the AUC@5° gap plausibly requires either training on multi-scene data with explicit hard-negative mining across scenes, or a stronger adapter family (e.g. joint LoRA on Q/K/V plus the FFN), both of which we leave to future work.

Author Contributions

The four team members contributed equally to this project, with a workload distribution of 25% each (totalling 100%).

Dai Xunlian (225040015, 25%) co-designed the overall recipe (LoRA on Q/V, HardInfoNCE, Safe Radius), implemented the LoRA injection module and the training loop, ran the ViT-S/16 and ViT-L/16 fine-tuning experiments, and co-drafted the Method and Experiments sections.

Mu Ruotong (225040014, 25%) built the NAVI data pipeline (scene-disjoint split, depth-based positive correspondence generation, training/test pair construction), implemented the HardInfoNCE loss and the 5-patch Safe-Radius mask, produced the per-epoch evaluation curves, and co-drafted the Dataset and Protocol subsections.

Li Langye (225040134, 25%) implemented the diagnostic analysis (intra-image / positive-pair cosine, effective rank, neighbourhood dominance, PCA-to-RGB visualization, Layer-4 pose probe), produced all diagnostic figures, and co-drafted the Diagnostic Analysis subsection together with the Limitations discussion.

Li Mingzhe (225045024, 25%) implemented the SuperGlue-style pose-evaluation pipeline (MNN, USAC, 5-pt with cheirality, AUC@5/10/20 and Precision metrics), managed the HuggingFace `Dinov3ViTModel` integration and the reproducibility scripts, and co-drafted the Introduction, Related Work and Conclusion sections.

All four authors jointly discussed the design decisions, reviewed each other's writing, and agree on the final version of the report.

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